

Домашнее задание 09 Гареевой Фариды

Репликация и удаление

Цель:

преобразовывать таблицу в реплицируемую;
настраивать реплики в ClickHouse и работать с данными в распределённой системе;

Описание/Пошаговая инструкция выполнения домашнего задания:

Для выполнения задания следуйте следующим шагам:

1. Возьмите любой демонстрационный DATASET: <https://clickhouse.com/docs/en/getting-started/example-datasets>.
2. Конвертируйте таблицу в реплицируемую, используя макрос replica.
3. Добавьте 2 реплики.
4. Выполните запросы и отдайте результаты как 2 файла:

```
SELECT  
getMacro('replica'),  
*  
FROM remote('разделенный запятыми список реплик',system.parts)  
FORMAT JSONEachRow;  
  
SELECT * FROM system.replicas FORMAT JSONEachRow;
```

5. Добавьте или выберите колонку с типом Date в таблице, добавьте TTL на таблицу «хранить последние 7 дней».

На проверку отправьте результат запроса «SHOW CREATE TABLE таблица»

Удачи при выполнении домашнего задания!

Если у вас возникнут вопросы, их всегда можно задать в чате ДЗ, нажав кнопку ниже, или в учебном Telegram канале группы

Критерии оценки:

Задание считается выполненным, если таблица конвертирована в реплицируемую, добавлены реплики, выполнены запросы с результатами в файлах и настроен TTL с проверкой через «SHOW CREATE TABLE».

Компетенции:

1. Масштабирование и оптимизация производительности
 - - знать методы горизонтального и вертикального масштабирования ClickHouse

- - настраивать масштабирование и управлять производительностью для обработки больших данных

ВЫПОЛНЕНИЕ ДОМАШНЕЙ РАБОТЫ

Возьмите любой демонстрационный

DATASET: <https://clickhouse.com/docs/en/getting-started/example-datasets>.

Я взяла Amazon Customer Review

У меня кликхаус развернут в докере и сразу загрузить данные через табличную функцию s3 не получается из-за слишком долгого скачивания. Я скачала файл на жесткий диск

https://datasets-documentation.s3.eu-west-3.amazonaws.com/amazon_reviews/amazon_reviews_2015.snappy.parquet

Подключила директорию из windows как volume и сделала перенос загруженного файла в директорию /etc/clickhouse-server/config.d/config.xml.

Развернула три сервера с кликхаус и один с Zookeeper в докере.

```
version: '3.8'

services:
  zookeeper:
    image: confluentinc/cp-zookeeper:latest
    container_name: zookeeper
    environment:
      ZOOKEEPER_CLIENT_PORT: 2181
      ZOOKEEPER_TICK_TIME: 2000
    ports:
      - 22181:2181
    networks:
      - clickhouse_net

  clickhouse-server-1:
    image: clickhouse/clickhouse-server:latest
    container_name: clickhouse-server-1
    hostname: clickhouse-server-1
    depends_on:
      - zookeeper
    ports:
      - "9000:9000"      # TCP для клиента
      - "8123:8123"      # HTTP интерфейс
      - "9009:9009"      # TCP для internal репликации
    volumes:
      - clickhouse_data-1:/var/lib/clickhouse # Хранение данных
      - clickhouse_config-1:/etc/clickhouse-server # Конфигурация
      (опционально)
      - c:\Users\gareeva\Downloads\mydata:/var/lib/clickhouse/files2 #
монтирование локальной директории в контейнер
    environment:
      CLICKHOUSE_USER: "click1"
      CLICKHOUSE_PASSWORD: "click1"
    networks:
      - clickhouse_net
```

```

clickhouse-server-2:
  image: clickhouse/clickhouse-server:latest
  container_name: clickhouse-server-2
  depends_on:
    - zookeeper
  ports:
    - "9002:9000"      # TCP для клиента
    - "8124:8123"      # HTTP интерфейс
    - "9010:9009"      # TCP для internal репликации
  environment:
    CLICKHOUSE_USER: "click2"
    CLICKHOUSE_PASSWORD: "click2"
  volumes:
    - clickhouse_data-2:/var/lib/clickhouse # Хранение данных
    - clickhouse_config-2:/etc/clickhouse-server # Конфигурация
(опционально)
  networks:
    - clickhouse_net

clickhouse-server-3:
  image: clickhouse/clickhouse-server:latest
  container_name: clickhouse-server-3
  depends_on:
    - zookeeper
  ports:
    - "9003:9000"      # TCP для клиента
    - "8125:8123"      # HTTP интерфейс
    - "9011:9009"      # TCP для internal репликации
  environment:
    CLICKHOUSE_USER: "click3"
    CLICKHOUSE_PASSWORD: "click3"
  volumes:
    - clickhouse_data-3:/var/lib/clickhouse # Хранение данных
    - clickhouse_config-3:/etc/clickhouse-server # Конфигурация
(опционально)
  networks:
    - clickhouse_net
volumes:
  clickhouse_data-1:
  clickhouse_data-2:
  clickhouse_data-3:
  clickhouse_config-1:
  clickhouse_config-2:
  clickhouse_config-3:
networks:
  clickhouse_net:

```

Зашла на каждый сервер кликхауса и в директории /etc/clickhouse-server/config.d/ создала конфигурационный файл config.xml (разница только в блоке macros -> replica, там значения replica-1, replica-2, replica-3 соответственно).

```

<clickhouse>
  <!-- Zookeeper -->
  <zookeeper>
    <node>
      <host>zookeeper</host>
      <port>2181</port>
    </node>
  </zookeeper>

```

```

<!-- Репликация -->
<remote_servers>
  <my_cluster>
    <shard>
      <replica>
        <host>clickhouse-server-1</host>
        <port>9000</port>
      </replica>
      <replica>
        <host>clickhouse-server-2</host>
        <port>9000</port>
      </replica>
      <replica>
        <host>clickhouse-server-3</host>
        <port>9000</port>
      </replica>
    </shard>
  </my_cluster>
</remote_servers>

<macros>
  <shard>my_shard</shard>
  <replica>replica-1</replica>
</macros>
</clickhouse>

```

На сервере clickhouse-server-1 создала таблицу (скопировала ddl из демонстрационных материалов на сайте)

```

create database amazon;

CREATE TABLE amazon.amazon_reviews
(
  `review_date` Date,
  `marketplace` LowCardinality(String),
  `customer_id` UInt64,
  `review_id` String,
  `product_id` String,
  `product_parent` UInt64,
  `product_title` String,
  `product_category` LowCardinality(String),
  `star_rating` UInt8,
  `helpful_votes` UInt32,
  `total_votes` UInt32,
  `vine` Bool,
  `verified_purchase` Bool,
  `review_headline` String,
  `review_body` String
)
ENGINE = MergeTree
ORDER BY (review_date, product_category);

```

Далее на клиенте попробовала загрузить файл

```
INSERT INTO amazon.amazon_reviews
```

```
SELECT *  
FROM file('amazon_reviews_2015.snappy.parquet', 'Parquet');
```

Сразу загрузить не получилось, размер файла более 8.6Гб, вылезла ошибка о недостаточности памяти. Увеличила доступную память серверу командой SET (настройку не помню).

Конвертируйте таблицу в реплицируемую, используя макрос replica.

Создала новую таблицу с той же структурой и движком ReplicatedMergeTree

```
CREATE TABLE amazon.amazon_reviews_replicated  
(  
    `review_date` Date,  
    `marketplace` LowCardinality(String),  
    `customer_id` UInt64,  
    `review_id` String,  
    `product_id` String,  
    `product_parent` UInt64,  
    `product_title` String,  
    `product_category` LowCardinality(String),  
    `star_rating` UInt8,  
    `helpful_votes` UInt32,  
    `total_votes` UInt32,  
    `vine` Bool,  
    `verified_purchase` Bool,  
    `review_headline` String,  
    `review_body` String  
)  
ENGINE =  
ReplicatedMergeTree('/clickhouse/{shard}/{database}/{table}', '{replica}')  
ORDER BY (review_date, product_category);
```

Добавьте 2 реплики.

На двух серверах clickhouse-server-2 и clickhouse-server-3 выполняю одинаковые запросы по созданию базы данных и реплицируемой таблицы:

```
CREATE TABLE amazon.amazon_reviews  
(  
    `review_date` Date,  
    `marketplace` LowCardinality(String),  
    `customer_id` UInt64,  
    `review_id` String,  
    `product_id` String,  
    `product_parent` UInt64,  
    `product_title` String,  
    `product_category` LowCardinality(String),  
    `star_rating` UInt8,  
    `helpful_votes` UInt32,  
    `total_votes` UInt32,  
    `vine` Bool,  
    `verified_purchase` Bool,  
    `review_headline` String,  
    `review_body` String  
)  
ENGINE =  
ReplicatedMergeTree('/clickhouse/{shard}/{database}/{table}', '{replica}')
```

```
ORDER BY (review_date, product_category);
```

На сервере clickhouse-server-1 перенесла данные из нереплицируемой таблицы `amazon.amazon_reviews` в реплицируемую `amazon.amazon_reviews_replicated` путем прикрепления партии `all`. Для этого сначала нашла все партии существующей таблицы следующим образом

```
select partition_id from system.parts where table='amazon_reviews';
```

Поскольку у нас таблица не секционирована, то у нас единственная партия `all`. Вот ее и приаттачим к новой таблице:

```
alter table amazon.amazon_reviews_replicated attach partition id 'all' from amazon.amazon_reviews;
```

Удаляем исходную таблицу:

```
drop table amazon.amazon_reviews sync;
```

Переименовываем реплицируемую таблицу

```
rename table amazon.amazon_reviews_replicated to amazon.amazon_reviews;
```

Выполните запросы и отдайте результаты как 2 файла:

```
SELECT getMacro('replica'),*  
FROM remote('clickhouse-server-1,clickhouse-server-2,clickhouse-server-3',  
'system.parts', 'click1', 'click1')  
WHERE table = 'amazon_reviews' FORMAT JSONEachRow;
```

```
{ "getMacro('replica')": "replica-  
1", "partition": "tuple()", "name": "all_10_10_0", "uuid": "00000000-0000-0000-  
0000-  
000000000000", "part_type": "Wide", "active": 1, "marks": "87", "rows": "702611", "by  
tes_on_disk": "156890601", "data_compressed_bytes": "156884689", "data_uncompres  
sed_bytes": "247522312", "primary_key_size": "512", "marks_bytes": "4368", "second  
ary_indices_compressed_bytes": "0", "secondary_indices_uncompressed_bytes": "0"  
, "secondary_indices_marks_bytes": "0", "modification_time": "2025-06-06  
21:12:19", "remove_time": "1970-01-01 00:00:00", "refcount": 1, "min_date": "1970-  
01-01", "max_date": "1970-01-01", "min_time": "1970-01-01  
00:00:00", "max_time": "1970-01-01  
00:00:00", "partition_id": "all", "min_block_number": "10", "max_block_number": "1  
0", "level": 0, "data_version": "10", "primary_key_bytes_in_memory": "0", "primary_  
key bytes in memory allocated": "0", "index granularity bytes in memory": "25",
```

```
"index_granularity_bytes_in_memory_allocated":"25","is_frozen":0,"database":
"amazon","table":"amazon_reviews","engine":"ReplicatedMergeTree","disk_name"
:"default","path":"\\var\\lib\\clickhouse\\store\\2d0\\2d0b5d3d-6b84-4093-
a55c-
e01d1214d3f8\\all_10_10_0\\","hash_of_all_files":"6f6faf7458051e87cc3bc576bc
52cd9b","hash_of_uncompressed_files":"f3a1fd1f37132c806e52dd7820c9b37a","unc
ompressed_hash_of_compressed_files":"7fc35e8e16e84f6522bcc0302983eab5","dele
te_ttl_info_min":"1970-01-01 00:00:00","delete_ttl_info_max":"1970-01-01
00:00:00","move_ttl_info.expression":[],"move_ttl_info.min":[],"move_ttl_inf
o.max":[],"default_compression_codec":"LZ4","recompression_ttl_info.expressi
on":[],"recompression_ttl_info.min":[],"recompression_ttl_info.max":[],"grou
p_by_ttl_info.expression":[],"group_by_ttl_info.min":[],"group_by_ttl_info.m
ax":[],"rows_where_ttl_info.expression":[],"rows_where_ttl_info.min":[],"row
s_where_ttl_info.max":[],"projections":[],"visible":1,"creation_tid":["1","1
","00000000-0000-0000-0000-
000000000000"],"removal_tid_lock":"0","removal_tid":["0","0","00000000-0000-
0000-0000-
000000000000"],"creation_csn":"1","removal_csn":"0","has_lightweight_delete"
:0,"last_removal_attempt_time":"1970-01-01
00:00:00","removal_state":"Cleanup thread hasn't seen this part yet"}
{"getMacro('replica')":"replica-
1","partition":"tuple()","name":"all_11_11_0","uuid":"00000000-0000-0000-
0000-
000000000000","part_type":"Wide","active":1,"marks":"38","rows":"297389","by
tes_on_disk":"68952212","data_compressed_bytes":"68947867","data_uncompresse
d_bytes":"108528823","primary_key_size":"305","marks_bytes":"3008","secondar
y_indices_compressed_bytes":"0","secondary_indices_uncompressed_bytes":"0","
secondary_indices_marks_bytes":"0","modification_time":"2025-06-06
21:12:19","remove_time":"1970-01-01 00:00:00","refcount":1,"min_date":"1970-
01-01","max_date":"1970-01-01","min_time":"1970-01-01
00:00:00","max_time":"1970-01-01
00:00:00","partition_id":"all","min_block_number":"11","max_block_number":"1
1","level":0,"data_version":"11","primary_key_bytes_in_memory":"0","primary_
key_bytes_in_memory_allocated":"0","index_granularity_bytes_in_memory":"25",
"index_granularity_bytes_in_memory_allocated":"25","is_frozen":0,"database":
"amazon","table":"amazon_reviews","engine":"ReplicatedMergeTree","disk_name"
:"default","path":"\\var\\lib\\clickhouse\\store\\2d0\\2d0b5d3d-6b84-4093-
a55c-
e01d1214d3f8\\all_11_11_0\\","hash_of_all_files":"29aefcf7aabe09967724750a45
158cc4","hash of uncompressed files":"8b021dae6dc7895efebac312aa301a98","unc
```

```
ompressed_hash_of_compressed_files":"2a05182ea527155a84f49c5a2de1bc7b","delete_ttl_info_min":"1970-01-01 00:00:00","delete_ttl_info_max":"1970-01-01 00:00:00","move_ttl_info.expression":[],"move_ttl_info.min":[],"move_ttl_info.max":[],"default_compression_codec":"LZ4","recompression_ttl_info.expression":[],"recompression_ttl_info.min":[],"recompression_ttl_info.max":[],"group_by_ttl_info.expression":[],"group_by_ttl_info.min":[],"group_by_ttl_info.max":[],"rows_where_ttl_info.expression":[],"rows_where_ttl_info.min":[],"rows_where_ttl_info.max":[],"projections":[],"visible":1,"creation_tid":["1","1","00000000-0000-0000-0000-000000000000"],"removal_tid_lock":"0","removal_tid":["0","0","00000000-0000-0000-000000000000"],"creation_csn":"1","removal_csn":"0","has_lightweight_delete":0,"last_removal_attempt_time":"1970-01-01 00:00:00","removal_state":"Cleanup thread hasn't seen this part yet"}{"getMacro('replica')":"replica-1","partition":"tuple()","name":"all_12_12_0","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Compact","active":1,"marks":"2","rows":"1","bytes_on_disk":"1692","data_compressed_bytes":"581","data_uncompressed_bytes":"206","primary_key_size":"61","marks_bytes":"107","secondary_indices_compressed_bytes":"0","secondary_indices_uncompressed_bytes":"0","secondary_indices_marks_bytes":"0","modification_time":"2025-06-06 21:19:39","remove_time":"1970-01-01 00:00:00","refcount":1,"min_date":"1970-01-01","max_date":"1970-01-01","min_time":"1970-01-01 00:00:00","max_time":"1970-01-01 00:00:00","partition_id":"all","min_block_number":"12","max_block_number":"12","level":0,"data_version":"12","primary_key_bytes_in_memory":"0","primary_key_bytes_in_memory_allocated":"0","index_granularity_bytes_in_memory":"25","index_granularity_bytes_in_memory_allocated":"25","is_frozen":0,"database":"amazon","table":"amazon_reviews","engine":"ReplicatedMergeTree","disk_name":"default","path":"/var/lib/clickhouse/store/2d0/2d0b5d3d-6b84-4093-a55c-e01d1214d3f8/all_12_12_0/","hash_of_all_files":"2b6fbbba7ff0b0a6a3eb582c214261b5","hash_of_uncompressed_files":"4da85e8c6a1c43d01660a75f99167b83","uncompressed_hash_of_compressed_files":"a4c76ccb03f87732b06625e093b18ecc","delete_ttl_info_min":"1970-01-01 00:00:00","delete_ttl_info_max":"1970-01-01 00:00:00","move_ttl_info.expression":[],"move_ttl_info.min":[],"move_ttl_info.max":[],"default_compression_codec":"LZ4","recompression_ttl_info.expression":[],"recompression_ttl_info.min":[],"recompression_ttl_info.max":[],"group_by_ttl_info.expression":[],"group by ttl info.min":[],"group by ttl info.m
```



```
ax":[],"rows_where_ttl_info.expression":[],"rows_where_ttl_info.min":[],"rows_where_ttl_info.max":[],"projections":[],"visible":1,"creation_tid":["1","1","00000000-0000-0000-0000-000000000000"],"removal_tid_lock":"0","removal_tid":["0","0","00000000-0000-0000-0000-000000000000"],"creation_csn":"1","removal_csn":"0","has_lightweight_delete":0,"last_removal_attempt_time":"1970-01-01 00:00:00","removal_state":"Cleanup thread hasn't seen this part yet"}{"getMacro('replica')":"replica-2","partition":"tuple()","name":"all_10_10_0","uuid":"00000000-0000-0000-0000-000000000000","part_type":"Wide","active":1,"marks":"87","rows":"702611","bytes_on_disk":"156890601","data_compressed_bytes":"156884689","data_uncompressed_bytes":"247522312","primary_key_size":"512","marks_bytes":"4368","secondary_indices_compressed_bytes":"0","secondary_indices_uncompressed_bytes":"0","secondary_indices_marks_bytes":"0","modification_time":"2025-06-06 21:12:22","remove_time":"1970-01-01 00:00:00","refcount":1,"min_date":"1970-01-01","max_date":"1970-01-01","min_time":"1970-01-01 00:00:00","max_time":"1970-01-01 00:00:00","partition_id":"all","min_block_number":"10","max_block_number":"10","level":0,"data_version":"10","primary_key_bytes_in_memory":"0","primary_key_bytes_in_memory_allocated":"0","index_granularity_bytes_in_memory":"25","index_granularity_bytes_in_memory_allocated":"25","is_frozen":0,"database":"amazon","table":"amazon_reviews","engine":"ReplicatedMergeTree","disk_name":"default","path":"\\var\\lib\\clickhouse\\store\\8e5\\8e54a7e0-1053-43d8-9c3f-4fad3ceee33f\\all_10_10_0\\","hash_of_all_files":"6f6faf7458051e87cc3bc576bc52cd9b","hash_of_uncompressed_files":"f3a1fd1f37132c806e52dd7820c9b37a","uncompressed_hash_of_compressed_files":"7fc35e8e16e84f6522bcc0302983eab5","delete_ttl_info_min":"1970-01-01 00:00:00","delete_ttl_info_max":"1970-01-01 00:00:00","move_ttl_info.expression":[],"move_ttl_info.min":[],"move_ttl_info.max":[],"default_compression_codec":"LZ4","recompression_ttl_info.expression":[],"recompression_ttl_info.min":[],"recompression_ttl_info.max":[],"group_by_ttl_info.expression":[],"group_by_ttl_info.min":[],"group_by_ttl_info.max":[],"rows_where_ttl_info.expression":[],"rows_where_ttl_info.min":[],"rows_where_ttl_info.max":[],"projections":[],"visible":1,"creation_tid":["1","1","00000000-0000-0000-0000-000000000000"],"removal_tid_lock":"0","removal_tid":["0","0","00000000-0000-0000-0000-000000000000"],"creation_csn":"1","removal_csn":"0","has_lightweight_delete"
```

```
:0,"last_removal_attempt_time":"1970-01-01
00:00:00","removal_state":"Cleanup thread hasn't seen this part yet"}
{"getMacro('replica')":"replica-
2","partition":"tuple()","name":"all_11_11_0","uuid":"00000000-0000-0000-
0000-
000000000000","part_type":"Wide","active":1,"marks":"38","rows":"297389","by
tes_on_disk":"68952212","data_compressed_bytes":"68947867","data_uncompresse
d_bytes":"108528823","primary_key_size":"305","marks_bytes":"3008","secondar
y_indices_compressed_bytes":"0","secondary_indices_uncompressed_bytes":"0","
secondary_indices_marks_bytes":"0","modification_time":"2025-06-06
21:12:22","remove_time":"1970-01-01 00:00:00","refcount":1,"min_date":"1970-
01-01","max_date":"1970-01-01","min_time":"1970-01-01
00:00:00","max_time":"1970-01-01
00:00:00","partition_id":"all","min_block_number":"11","max_block_number":"1
1","level":0,"data_version":"11","primary_key_bytes_in_memory":"0","primary
key_bytes_in_memory_allocated":"0","index_granularity_bytes_in_memory":"25",
"index_granularity_bytes_in_memory_allocated":"25","is_frozen":0,"database":
"amazon","table":"amazon_reviews","engine":"ReplicatedMergeTree","disk_name"
:"default","path":"\\var\\lib\\clickhouse\\store\\8e5\\8e54a7e0-1053-43d8-
9c3f-
4fad3ceee33f\\all_11_11_0\\","hash_of_all_files":"29aelcf7aabe09967724750a45
158cc4","hash_of_uncompressed_files":"8b021dae6dc7895efebac312aa301a98","unc
ompressed_hash_of_compressed_files":"2a05182ea527155a84f49c5a2de1bc7b","dele
te_ttl_info_min":"1970-01-01 00:00:00","delete_ttl_info_max":"1970-01-01
00:00:00","move_ttl_info.expression":[],"move_ttl_info.min":[],"move_ttl_inf
o.max":[],"default_compression_codec":"LZ4","recompression_ttl_info.expressi
on":[],"recompression_ttl_info.min":[],"recompression_ttl_info.max":[],"grou
p_by_ttl_info.expression":[],"group_by_ttl_info.min":[],"group_by_ttl_info.m
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```

```
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```

```
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y_indices_compressed_bytes":"0","secondary_indices_uncompressed_bytes":"0","
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21:12:22","remove_time":"1970-01-01 00:00:00","refcount":1,"min_date":"1970-
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00:00:00","max_time":"1970-01-01
00:00:00","partition_id":"all","min_block_number":"11","max_block_number":"1
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"index_granularity_bytes_in_memory_allocated":"25","is_frozen":0,"database":
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```

```
: "default", "path": "\\var\\lib\\clickhouse\\store\\dd5\\dd58fc79-dc6e-4c11-98d1-d290b2a01241\\all_11_11_0\\", "hash_of_all_files": "29aefcf7aabe09967724750a45158cc4", "hash_of_uncompressed_files": "8b021dae6dc7895efebac312aa301a98", "uncompressed_hash_of_compressed_files": "2a05182ea527155a84f49c5a2de1bc7b", "delete_ttl_info_min": "1970-01-01 00:00:00", "delete_ttl_info_max": "1970-01-01 00:00:00", "move_ttl_info.expression": [], "move_ttl_info.min": [], "move_ttl_info.max": [], "default_compression_codec": "LZ4", "recompression_ttl_info.expression": [], "recompression_ttl_info.min": [], "recompression_ttl_info.max": [], "group_by_ttl_info.expression": [], "group_by_ttl_info.min": [], "group_by_ttl_info.max": [], "rows_where_ttl_info.expression": [], "rows_where_ttl_info.min": [], "rows_where_ttl_info.max": [], "projections": [], "visible": 1, "creation_tid": ["1", "1", "00000000-0000-0000-0000-000000000000"], "removal_tid_lock": "0", "removal_tid": ["0", "0", "00000000-0000-0000-000000000000"], "creation_csn": "1", "removal_csn": "0", "has_lightweight_delete": 0, "last_removal_attempt_time": "1970-01-01 00:00:00", "removal_state": "Cleanup thread hasn't seen this part yet"}
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```

```
00:00:00","move_ttl_info.expression":[],"move_ttl_info.min":[],"move_ttl_info.max":[],"default_compression_codec":"LZ4","recompression_ttl_info.expression":[],"recompression_ttl_info.min":[],"recompression_ttl_info.max":[],"group_by_ttl_info.expression":[],"group_by_ttl_info.min":[],"group_by_ttl_info.max":[],"rows_where_ttl_info.expression":[],"rows_where_ttl_info.min":[],"rows_where_ttl_info.max":[],"projections":[],"visible":1,"creation_tid":["1","1","00000000-0000-0000-0000-000000000000"],"removal_tid_lock":"0","removal_tid":["0","0","00000000-0000-0000-0000-000000000000"],"creation_csn":"1","removal_csn":"0","has_lightweight_delete":0,"last_removal_attempt_time":"1970-01-01 00:00:00","removal_state":"Cleanup thread hasn't seen this part yet"}
```

```
SELECT * FROM system.replicas FORMAT JSONEachRow;
```

```
{"database":"amazon","table":"amazon_reviews","engine":"ReplicatedMergeTree","is_leader":1,"can_become_leader":1,"is_readonly":0,"readonly_start_time":null,"is_session_expired":0,"future_parts":0,"parts_to_check":0,"zookeeper_name":"default","zookeeper_path":"\\clickhouse\\my_shard\\amazon\\amazon_reviews_replicated","replica_name":"replica-1","replica_path":"\\clickhouse\\my_shard\\amazon\\amazon_reviews_replicated\\replicas\\replica-1","columns_version":1,"queue_size":0,"inserts_in_queue":0,"merges_in_queue":0,"part_mutations_in_queue":0,"queue_oldest_time":"1970-01-01 00:00:00","inserts_oldest_time":"1970-01-01 00:00:00","merges_oldest_time":"1970-01-01 00:00:00","part_mutations_oldest_time":"1970-01-01 00:00:00","oldest_part_to_get":"","oldest_part_to_merge_to":"","oldest_part_to_mutate_to":"","log_max_index":"8","log_pointer":"9","last_queue_update":"1970-01-01 00:00:00","absolute_delay":"0","total_replicas":3,"active_replicas":3,"lost_part_count":"2","last_queue_update_exception":"","zookeeper_exception":"","replica_is_active":{"replica-1":1,"replica-2":1,"replica-3":1}}
```

Добавьте или выберите колонку с типом Date в таблице, добавьте TTL на таблицу «хранить последние 7 дней».

На проверку отправьте результат запроса «SHOW CREATE TABLE таблица»

```
ALTER TABLE amazon.amazon_reviews
MODIFY TTL review_date + INTERVAL 7 DAY;
```

```
show create table amazon.amazon_reviews format TSVRaw;
```

Результат:

```
CREATE TABLE amazon.amazon_reviews
```

```
(
    `review_date` Date,
    `marketplace` LowCardinality(String),
    `customer_id` UInt64,
    `review_id` String,
    `product_id` String,
    `product_parent` UInt64,
    `product_title` String,
    `product_category` LowCardinality(String),
    `star_rating` UInt8,
    `helpful_votes` UInt32,
    `total_votes` UInt32,
    `vine` Bool,
    `verified_purchase` Bool,
    `review_headline` String,
    `review_body` String
)
ENGINE = ReplicatedMergeTree('/clickhouse/{shard}/amazon/amazon_reviews_replicated',
'{replica}')
ORDER BY (review_date, product_category)
TTL review_date + toIntervalDay(7)
SETTINGS index_granularity = 8192;
```